

Abstract Algebra [parts I, II] – Dummit & Foote

Notes by Gene

2026

Contents

I	Group Theory	3
1	Introduction to Groups	3
1.1	Axioms	3
1.2	Dihedral Groups	3
1.3	Symmetric Groups	4
1.4	Matrix Groups	4
1.5	The Quaternion Group	5
1.6	Homomorphisms and Isomorphisms	5
1.7	Group Actions	5
2	Subgroups	6
2.1	Centralizers and Normalizers, Stabilizers and Kernels	6
2.2	Cyclic Subgroups	7
2.2.1	Theorems	7
2.3	Subgroups Generated by Subsets	8
3	Quotient Groups and Homomorphisms	9
3.1	Definitions	9
3.2	Theorems	10
3.2.1	Quotient Groups from Kernel	10
3.2.2	Quotient Groups from Normal Subgroups	11
3.3	Lagrange's Theorem	12
3.3.1	Statement	12
3.3.2	Corollaries	12
3.3.3	Cauchy's Theorem and Sylow's Theorem	12
3.3.4	Applications: Fermat's Little Theorem, Euler's Theorem	13
3.4	Isomorphism Theorems	13
3.5	The Hölder Program	14
3.5.1	Composition Series	14
3.5.2	Hölder Program	14
3.6	The Alternating Group A_n	14
4	Group Actions	15

4.1	Definitions	15
4.2	Groups Acting on Themselves by Conjugation	16
4.2.1	Definitions	16
4.2.2	Conjugacy in Symmetric Groups	17
4.3	Automorphisms	17
4.4	Sylow Theorems	18
4.4.1	Applications	19
5	Direct and Semidirect Products and Abelian Groups	19
5.1	Direct Products	19
5.2	Fundamental Theorem of Finitely-Generated Abelian Groups	20
5.3	Recognizing Direct Products	21
II	Ring Theory	21
6	Introduction to Rings	22
6.1	Definitions	22
6.2	Types of Rings	23
6.2.1	Polynomial Rings	23
6.2.2	Matrix Rings	23
6.2.3	Group Rings	23
6.3	Ring Homomorphisms	23
6.3.1	Ring Ideals	24
6.3.2	The Isomorphism Theorems	24
6.3.3	Solving Diophantine Equations with $\phi : \mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$	25
6.4	Properties of Ideals	25
6.4.1	Finitely Generated Ideals	25
6.4.2	Maximal Ideals	25
6.4.3	Prime Ideals	26
6.5	Rings of Fractions	26
6.6	Chinese Remainder Theorem	27
7	Domains, Domains, Domains	27
7.1	Euclidean Domains	27
7.1.1	Definitions	27
7.1.2	Greatest Common Divisors	28
7.2	Principal Ideal Domains	28
7.2.1	Dedekind-Hasse Norm	29
7.3	Unique Factorization Domains	29
7.3.1	Definitions	29
7.3.2	Fermat's Theorem on Sums of Squares	30
8	Polynomial Rings	30
8.1	Definitions	30
8.1.1	Polynomial Rings Over Fields	31
8.2	Factoring Polynomials	32
8.3	Gröbner Bases	32

Part I

Group Theory

1 Introduction to Groups

1.1 Axioms

If $\star$ is a function on set G such that $\star : G \times G \rightarrow G$, then $\star$ is a **binary operation** on G .

- Associative if $\forall a, b \in G, a \star (b \star c) = (a \star b) \star c$
- Commutative if $\forall a, b \in G, a \star b = b \star a$

A **group** is an ordered pair $(G, \star)$ such that

1. $\star$ is associative
2. Existence of identity: $\exists e \in G$ such that $\forall a \in G, a \star e = e \star a = a$
3. Existence of inverse: $\forall a \in G, \exists a^{-1} \in G$ such that $a \star a^{-1} = a^{-1} \star a = e$

*A group is **Abelian (commutative)** if $\forall a, b \in G, a \star b = b \star a$

E.g. $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$ are groups under $+$ with identity $e = 0$ and inverse $a^{-1} = -a$. But $\mathbb{Z} - \{0\}$ is not a group under $\times$ because inverses are not guaranteed to exist.

Let $(G, \star)$ be a group, and $x \in G$. The **order** of x is the smallest positive integer n such that $x^n = 1$, and is denoted $|x|$. If no positive powers of x is 1, $|x| = \infty$.

- In the additive group $\mathbb{Z}/9\mathbb{Z}$, $\bar{6}$ has order 3, since $\bar{6} + \bar{6} + \bar{6} = \bar{18} = \bar{0}$.
- In the multiplicative group $\mathbb{R} - \{0\}$, -1 has order 2 since $(-1)^2 = 1$.

1.2 Dihedral Groups

For each $n \in \mathbb{Z}^+, n \geq 3$, let D_{2n} be the set of symmetries of a regular n -gon. E.g. $s \in D_{2n}$ might be a rotation of $2\pi/n$ around the n -gon.

$|D_{2n}| = 2n$, so we call D_{2n} the **dihedral group of order $2n$** .

Any symmetry can be written in terms of rotations (r) and reflections (s) alone, e.g. for $n = 12$,

$$(sr^9)(sr^6) = s(r^9s)r^6 = s(sr^{-9})r^6 = s^2r^{-3} = r^9$$

S is a set of generators of G if any element of G can be written as a finite product of the elements of S . This is notated:

$$G = \langle S \rangle$$

For the dihedral group, $S = \{s, r\}$. The equations in G that the generators satisfy are called **relations** in G . For D_{2n} , the relations are $r^n = 1$, $s^2 = 1$, and $rs = sr^{-1}$.

If G is generated by S , and there exists a collection of relations $R_1, R_2, \dots, R_m$, then these generators and relations are called the **presentation of G** :

$$G = \langle S \mid R_1, R_2, \dots, R_m \rangle$$

For the dihedral group, $D_{2n} = \langle r, s \mid r^n = s^2 = 1, rs = sr^{-1} \rangle$.

1.3 Symmetric Groups

Let S_Ω be the bijection from Ω to itself (permutation). S_Ω is a group under the function composition $\circ$.

- Call $(S_\Omega, \circ)$ the **symmetric group** on set Ω .
- If $\Omega = \{1, 2, 3, \dots, n\}$, then we denote S_Ω as S_n , the **symmetric group of degree n** .
- $(S_n, \circ)$ has degree $n!$ ($n!$ elements).

We can represent S_n in a cycle decomposition, grouping disjoint cycles. E.g.

$$(a_1 \ a_2 \ \dots \ a_m)(a_{m_1+1} \ a_{m_1+2} \ \dots \ a_{m_2}) \cdots (a_{m_{k-1}+1} \ a_{m_{k-1}+2} \ \dots \ a_{m_k})$$

Meaning that a_1 goes to a_2 , a_{m_1+1} goes to a_{m_1+2} , etc. Values such that $\sigma(a_i) = a_i$ aren't written. For example, for $\sigma \in S_3$ with $\sigma(1) = 1$, $\sigma(2) = 3$, and $\sigma(3) = 2$, the cycle decomposition is

$$(2 \ 3)$$

1.4 Matrix Groups

First, we need to define a field. A field is a set F with 2 binary operators $+$ and $\cdot$ such that

1. $(F, +)$ and $(F - \{0\}, \cdot)$ are Abelian groups
2. The distribution law holds:

$$a \cdot (b + c) = (a \cdot b) + (a \cdot c) \quad \forall a, b, c \in F$$

Define the **general linear group of degree n** :

$$GL_n(F) = \{A \mid A \text{ is a } n \times n \text{ matrix with entries from } F, \text{ and } \det(A) \neq 0\}$$

- Since $\det(A) = 0$, $GL_n(F)$ is closed under matrix multiplication.
- Matrix products are associative.
- $\det(A) \neq 0$ iff. A has an inverse.

Thus, $GL_n(F)$ is a group under matrix multiplication.

1.5 The Quaternion Group

Simple enough:

$$\begin{aligned}
 Q_8 &= \{1, -1, i, -i, j, -j, k, -k\} \\
 1 \cdot a &= a \cdot 1 = a \\
 (-1) \cdot a &= a \cdot (-1) = -a \\
 i \cdot i &= j \cdot j = k \cdot k = -1 \\
 i \cdot j &= k, \quad j \cdot k = i, \quad k \cdot i = j \\
 j \cdot i &= -k, \quad k \cdot j = -i, \quad i \cdot k = -j
 \end{aligned}$$

$(Q_8, \cdot)$ is a non-Abelian group of order 8.

1.6 Homomorphisms and Isomorphisms

Let $(G, \star)$ and $(H, \diamond)$ be groups. A map $\phi : G \rightarrow H$ such that

$$\phi(x \star y) = \phi(x) \diamond \phi(y)$$

is a **homomorphism**.

- “A map that reflects the group structure of the domain and codomain.”

$\phi : G \rightarrow H$ is called an **isomorphism** if

1. ϕ is a homomorphism
2. ϕ is a bijection

Can also say: G and H are isomorphic; G and H are of the same isomorphism type; $G \cong H$.

I.e. G and H are the “same group” except the elements and operators are written differently (think abstract).

If $\phi : G \rightarrow H$ is an isomorphism, then

1. $|G| = |H|$
2. G is Abelian iff. H is Abelian
3. $\forall x \in G, |x| = |\phi(x)|$

1.7 Group Actions

A group action from group G on set A is a map from $G \times A$ to A satisfying

1. $g_1 \cdot (g_2 \cdot a) = (g_1 g_2) \cdot a \quad \forall g_1, g_2 \in G, \forall a \in A$
 2. $1 \cdot a = a \quad \forall a \in A$
-

Let G act on A . For each $g \in G$, define σ_g to be the map

$$\begin{aligned}\sigma_g : A &\rightarrow A \\ \sigma_g(a) &= g \cdot a\end{aligned}$$

1. For each $g \in G$, σ_g is a permutation of A .
2. The map from G to S_A defined by $g \mapsto \sigma_g$ is a homomorphism.

I.e. every $g \in G$ acts as a permutation of A in a way consistent with the group operation in G .

Group actions treat $g \in G$ to be functions, sort of.

2 Subgroups

A subset H of a group H is a **subgroup** of G if

1. $H \neq \emptyset$
2. H is closed under products and inverses. I.e. $x, y \in H$ implies $x^{-1} \in H$ and $xy \in H$.

This is denoted $H \leq G$, or if $H \neq G$, $H < G$.

I.e. a subgroup is a subset of G that is also a group w.r.t the binary operation defined on G . A subgroup must thus contain the identity operator.

Every G has two subgroups: $H = G$ and $H = \{1\}$ (trivial subgroup).

A simpler subgroup criterion: a subset H of a group G is a subgroup of G iff.

1. $H \neq \emptyset$
2. $\forall x, y \in H, xy^{-1} \in H$

2.1 Centralizers and Normalizers, Stabilizers and Kernels

Let A be a subset of G .

The **centralizer of A in G** is the set of elements of G that commute with every element of A .

$$C_G(A) = \{g \in G \mid gag^{-1} = a, \forall a \in A\}$$

- For an Abelian group G , $C_G(A) = G, \forall A \subseteq G$.
- For the quaternion group, $C_{Q_8}(\{i\}) = \{\pm 1, \pm i\}$

The **center of G** is the set of elements of G commuting with every element of G .

$$Z(G) = C_G(G) = \{g \in G \mid gx = xg, \forall x \in G\}$$

The **normalizer of A in G** is the set of elements of G that commutes with the entire set A .

$$N_G(A) = \{g \in G \mid gAg^{-1} = A\}$$

This is a looser criterion than the centralizer, since order within A doesn't matter. Formally, if $g \in C_G(A)$, then $gag^{-1} = a$ for all $a \in A$, so $C_G(A) \leq N_G(A)$.

For an Abelian group G , which has the commutation property already, $C_G(A) = N_G(A) = Z(G) = G$, for any subset $A \subseteq G$.

If G is a group action on set S , and $s \in S$, the **stabilizer of s in G** is

$$G_s = \{g \in G \mid g \cdot s = s\}$$

- $1 \in G_s$
- If $y \in G_s$, then $y^{-1} \in G_s$.

$$\begin{aligned} s &= 1 \cdot s = (y^{-1}y) \cdot s \\ &= y^{-1} \cdot (y \cdot s) \\ &= y^{-1} \cdot s \end{aligned}$$

- If $x, y \in G_s$, then $xy \in G_s$.

$$\begin{aligned} (xy) \cdot s &= x \cdot (y \cdot s) \\ &= x \cdot s \\ &= s \end{aligned}$$

Thus G_s is a subgroup of the group action G , or $G_s \leq G$.

The **kernel of the action of G on set S** is

$$\{g \in G \mid g \cdot s = s, \forall s \in S\}$$

2.2 Cyclic Subgroups

A possible way to generate a subgroup H of G from an element $x \in G$ is to form the set $H = \{x^n \mid n \in \mathbb{Z}\}$. Guarantees closure under products and inverses.

A subgroup $H \subseteq G$ is **cyclic** if $\exists x \in H$ such that $H = \{x^n \mid n \in \mathbb{Z}\}$. I.e. H can be generated by 1 element. Notated $H = \langle x \rangle$. Cyclic groups are Abelian, by properties of the exponent.

- E.g. If $H = (\mathbb{Z}, +)$, then $H = \langle 1 \rangle$.
- E.g. If H is the subgroup of all rotations of an n -gon, then $H = \langle r \rangle$.

2.2.1 Theorems

1. If $H = \langle x \rangle$, then $|H| = |x|$.
2. Let $x \in G$. For $m, n \in \mathbb{Z}$, if $x^n = x^m = 1$, then $x^d = 1$, where $d = \gcd(m, n)$.
3. Any 2 cyclic groups of the same order are isomorphic.

- If $n \in \mathbb{Z}^+$ and $\langle x \rangle, \langle y \rangle$ are cyclic groups of order n , the following map is an isomorphism.

$$\begin{aligned}\phi : \langle x \rangle &\rightarrow \langle y \rangle \\ x^k &\mapsto y^k\end{aligned}$$

- If $\langle x \rangle$ is an infinite cyclic group, the following map is a homomorphism. Thus for each $n \in \mathbb{Z}^+$, let Z_n be the cyclic group of order n .

$$\begin{aligned}\phi : \mathbb{Z} &\rightarrow \langle x \rangle \\ k &\mapsto x^k\end{aligned}$$

4. Let G be a map, $x \in G$, and $a \in \mathbb{Z} - \{0\}$.

- If $|x| = \infty$, then $|x^a| = \infty$
- If $|x| = n < \infty$, then $|x^a| = n / \gcd(n, a)$

5. Let $H = \langle x \rangle$.

- If $|x| = \infty$, then $H = \langle x \rangle$ iff. $a = \pm 1$.
- If $|x| = n < \infty$, then $H = \langle x^a \rangle$ iff. $\gcd(a, n) = 1$. The number of generators of H is $\phi(n)$.
- E.g. $\bar{a}$ generates $\mathbb{Z}/n\mathbb{Z}$ iff. $(\bar{a}, n) = 1$. So $\bar{1}, \bar{5}, \bar{7}, \bar{11}$ are generators of $\mathbb{Z}/12\mathbb{Z}$, and these are all the $\phi(12) = 4$ generators.

6. Let $H = \langle x \rangle$ be a cyclic group. Then

- Every subgroup of H is cyclic.
- If $|H| = \infty$, then for any non-negative and distinct $a, b \in \mathbb{Z}$, $\langle x^a \rangle = \langle x^b \rangle$.
- If $|H| = n < \infty$, then for each $a \in \mathbb{Z}^+$ such that $a|n$, there's a unique subgroup of order a . This is the cyclic group $\langle x^d \rangle$, where $d = n/a$.

2.3 Subgroups Generated by Subsets

If $\mathcal{A}$ is any nonempty collection of subgroups of G , the intersection of all $\mathcal{A}$ is also a subgroup of G .

$$K = \bigcap_{H \in \mathcal{A}} H \quad \text{is a group if} \quad H \leq G$$

If A is a subset of the group G , let the **subgroup of G generated by A** be

$$\langle A \rangle = \bigcap_{A \subseteq H, H \leq G} H$$

I.e. the intersection of all subgroups of G containing A . Equivalently,

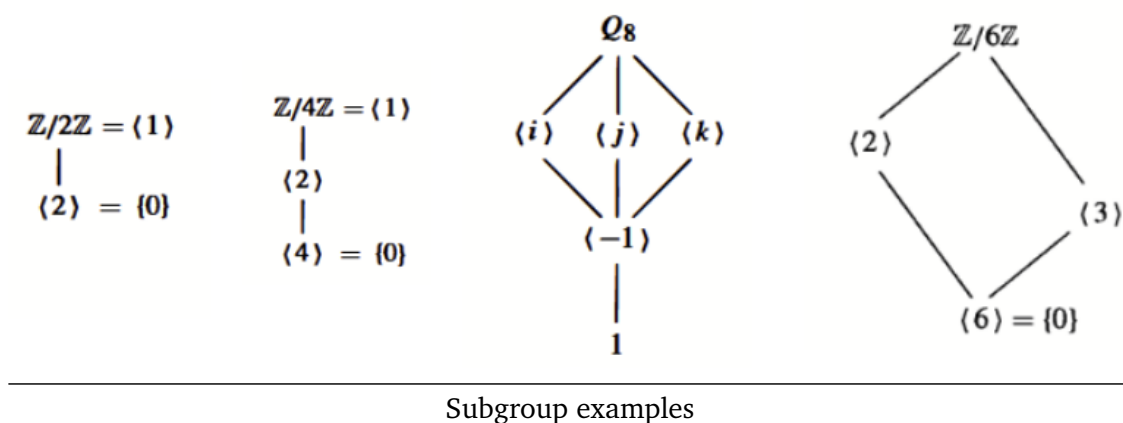
$$\langle A \rangle = \{a_1^{\alpha_1} a_2^{\alpha_2} \cdots a_n^{\alpha_n} \mid \forall i, a_i \in A, \alpha_i \in \mathbb{Z}, a_i \neq a_{i+1} \text{ and } n \in \mathbb{Z}^+\}$$

An Abelian group G simplifies this a bit:

$$\langle A \rangle = \{a_1^{\alpha_1} a_2^{\alpha_2} \cdots a_n^{\alpha_n} \mid \forall i \text{ and } \alpha_i \in \mathbb{Z}\}$$

- Without an Abelian structure, determining the generated subgroup becomes difficult. E.g. if $G = D_8$, let r, s be the generators. Let $a = s, b = rs$ so that $G = \langle a, b \rangle$. Thus G is generated by 2 elements of order 2, but has order 8.
- Or for $G = S_n = \langle (1\ 2), (1\ 2\ 3\ \dots\ n) \rangle$. Generated by elements of order 2 and n , but has order $n!$

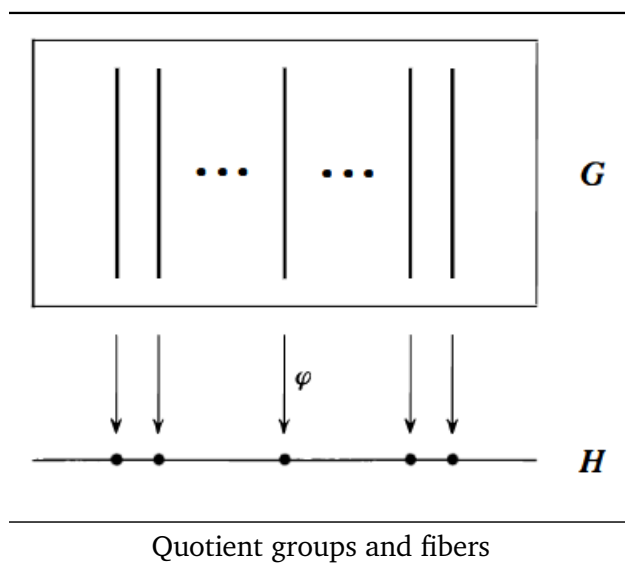
Lattice diagrams can help illuminate the structure of groups and subgroups. Build up from $\{0\}$ at the bottom, and draw subgroups of increasingly higher order upwards. This is helpful to see, for example, the smallest subgroup of G , $\langle H, K \rangle$, containing two smaller subgroups $H \leq G$ and $K \leq G$.



3 Quotient Groups and Homomorphisms

3.1 Definitions

Quotient groups are a way to obtain a smaller group from a group G .



$\phi : G \rightarrow H$ is a homomorphism from G to H , i.e. $\phi(xy) = \phi(x)\phi(y)$ for $x, y \in G$. Vertical lines above each point $h \in H$ represent the **fiber of ϕ over h** .

Denote the **kernel of ϕ** as the set

$$\ker \phi = \{g \in G \mid \phi(g) = 1_H\}$$

Let K be the kernel. The **quotient group/factor group** G/K (" $G \bmod K$ ") is a group whose elements are the fibers of ϕ .

For any subgroup $N \leq G$ and $g \in G$, define

$$\text{Left coset of } N \text{ in } G: gN = \{gn \mid n \in N\}$$

$$\text{Right coset of } N \text{ in } G: Ng = \{ng \mid n \in N\}$$

Call any element of a coset the **representative** of that coset. The fibers of a homomorphism $\phi : G \rightarrow H$ are the left or right cosets of the kernel (it may help to think of this graphically). The elements of G/K are the left cosets gK or the right cosets Kg with $g \in G$.

3.2 Theorems

3.2.1 Quotient Groups from Kernel

Theorem 1: Let $\phi : G \rightarrow H$ be a homomorphism. Then

1. $\phi(1_G) = 1_H$
2. $\phi(g^{-1}) = \phi(g)^{-1}$
3. $\phi(g^n) = \phi(g)^n$
4. Kernel of ϕ is a subgroup of G : $\ker \phi \leq G$
5. Image of G under ϕ is a subgroup of H : $\text{im}(\phi) \leq H$

Proof: For (1), $\phi(1_G) = \phi(1_G 1_G) = 1_H 1_H$. For (2), $\phi(g)\phi(g^{-1}) = \phi(gg^{-1}) = \phi(1_G) = 1_H$. Then (3) follows from induction.

For (4), note that $1_G \in \ker \phi$, so $\ker \phi$ is nonempty. Let $x, y \in \ker \phi$. Then $\ker \phi$ satisfies the subgroup criterion:

$$\phi(xy^{-1}) = \phi(x)\phi(y)^{-1} = 1_H$$

Similarly for (5), note that $\phi(1_G) = 1_H$, so the identity of H lies in $\text{im}(\phi)$. If $x, y \in \text{im}(\phi)$ with $\phi(a) = x$ and $\phi(b) = y$, then $\text{im}(\phi)$ satisfies the subgroup criterion:

$$\begin{aligned} xy^{-1} &= \phi(a)\phi(b)^{-1} = \phi(ab^{-1}) \\ xy^{-1} &\in \text{im}(\phi) \end{aligned}$$

Definition: Let $\phi : G \rightarrow H$ with kernel K . Then the **quotient group** G/K is the group whose elements are fibers of ϕ with group operations defined such that: if X is the fiber above a and Y is the fiber above b , then $X \cdot Y$ is the fiber above ab .

Thus $K \in G/K$. Other elements of G/K are “translations” of K .

Theorem 2: Let $\phi : G \rightarrow H$ be a homomorphism with kernel K . Let $X \in G/K$ be a fiber above a , i.e. $X = \phi^{-1}(a)$. Then

- 1) $\forall u \in X, X = \{uk \mid k \in K\}$
- 2) $\forall u \in X, X = \{ku \mid k \in K\}$

Proof: First show $uK \subseteq X$, then $X \subseteq uK$. Similarly for Ku .

What this shows is that the fibers of a homomorphism are the left (or right) cosets of the kernel. Furthermore, the specific value of u doesn't matter, as long as $u \in X$. This is why u is called the “**representative**.”

Theorem 3: Let K be the kernel of $\phi : G \rightarrow H$. The set of left cosets of G with operation defined as

$$uK \cdot vK = (uv)K$$

form the quotient group G/K . Similarly for right cosets.

E.g. for $\phi : \mathbb{Z} \rightarrow \mathbb{Z}_n$, the cosets are $a + n\mathbb{Z}$ and the kernel is $n\mathbb{Z}$. Thus the fibers $\mathbb{Z}/n\mathbb{Z}$ form a group.

In essence, to find a quotient group:

1. Find homomorphism ϕ from group G .
2. Find kernel K of homomorphism.
3. G/K , the fibers of ϕ , is a group

3.2.2 Quotient Groups from Normal Subgroups

We can form a group from a subgroup N iff. N is the kernel of some homomorphism. To determine whether N is a kernel of such a homomorphism, check whether N is a normal subgroup: $N \trianglelefteq G$.

Theorem: For a subgroup N of G , i.e. $N \leq G$, the following are equivalent conditions.

1. $N \trianglelefteq G$
2. $N_G(N) = G$
3. $gN = Ng, \quad \forall g \in G$
4. $gNg^{-1} \subseteq N, \quad \forall g \in G$

Theorem: $N \leq G$ is normal iff. it's the kernel of some homomorphism.

Proof: If N is the kernel of some homomorphism, its left and right cosets are equal (both are fibers of ϕ). By the previous theorem, if $gN = Ng$ for all $g \in G$ is equivalent to $N \trianglelefteq G$.

In the opposite direction, assume $N \trianglelefteq G$ and let $\pi : G \rightarrow G/N$ with $\pi(g) = gN$. Then

$$\pi(g_1g_2) = g_1g_2N = (g_1N)(g_2N) = \pi(g_1)\pi(g_2)$$

so π is a homomorphism. The kernel is $\ker \pi = \{g \in G \mid gN = 1N\} = N$, the normal subgroup.

For $N \trianglelefteq G$, the homomorphism $\pi : G \rightarrow G/N$ defined by $\pi(g) = gN$ is the **natural projection homomorphism**.

Thus an easy way to see whether a subgroup $N \leq G$ is a kernel of some homomorphism is to check whether $N_g(N) = G$. Then we can find the quotient group G/N .

Note that in an Abelian group, any subgroup $N \leq G$ is normal, so any $N \leq G$ can be used to create a quotient group G/N .

3.3 Lagrange's Theorem

3.3.1 Statement

Theorem: The left cosets of $N \leq G$ form a partition of G , and $uN = vN$ iff. u, v are representatives of the same coset.

Lagrange's Theorem: If G is a finite group and $H \leq G$, then $|H|$ divides $|G|$. The number of left cosets of H in G is $|G|/|H|$.

Proof: Let $|H| = n$ and k be the number of left cosets of H in G . For some $g \in G$, define $\phi : H \rightarrow gH$ with $\phi(h) = gh$. ϕ is both injective and surjective, so $|gH| = |H| = n$. From the previous theorem, the cosets partition G , so $|G| = k|H| = kn$. Thus $|G|/|H| = k$.

If $H \leq G$, the number of left cosets of H in G is called the **index** of H in G , denoted $|G : H|$.

- E.g. $\{0\} \in \mathbb{Z}$ has infinite index.
- E.g. $\langle n \rangle \in \mathbb{Z}$ has index n .

3.3.2 Corollaries

Corollary 1: If $x \in G$ and $|G|$ is finite, then $|x|$ divides $|G|$, i.e. $x^{|G|} = 1$.

Proof: $|x| = |\langle x \rangle|$. Let $H = \langle x \rangle$. Then $H \leq G$ so $|H| = |x|$ divides $|G|$.

Corollary 2: If $|G|$ is prime, then G is cyclic and $G \cong Z_p$.

Proof: Let $x \in G$ and $x \neq 1$. Then $|\langle x \rangle| \geq 1$. $\langle x \rangle \in G$, so $|\langle x \rangle|$ divides $|G|$. But $|G|$ is prime, so $|G| = |\langle x \rangle|$. Hence G is cyclic with order p .

Corollary 3: Let $H \leq G$ and $|G : H| = 2$. Then $H \trianglelefteq G$.

Proof: Let $g \in G - H$. The two left cosets of H in G are $1H$ and gH . They partition G , so $gH = G - H$. Similarly, $Hg = g - H$. Then $gH = Hg$, so H must be a normal subgroup of G .

- E.g. $\langle i \rangle = \{i, -1, -i, +1\}$, which has order 4. Q_8 has order 8, so $|Q_8 : \langle i \rangle| = 2$. Thus $\langle i \rangle \trianglelefteq Q_8$.

3.3.3 Cauchy's Theorem and Sylow's Theorem

The converse to Lagrange's theorem isn't true generally: just because n divides G doesn't mean there exists $H \leq G$ with $|H| = n$.

There are some partial converses:

Cauchy's Theorem: If G is finite, and there exists a prime p dividing $|G|$, then G has an element of order p .

Sylow's Theorem: If G is a finite group of order $p^\alpha m$, where p is prime and doesn't divide m , then G has a subgroup of order p^α .

3.3.4 Applications: Fermat's Little Theorem, Euler's Theorem

Lagrange's theorem provides an elegant way to prove these (seemingly number-theoretic) theorems.

Fermat's Little Theorem: If p is prime, then $a^p \equiv a \pmod{p}$ for all $a \in \mathbb{Z}$.

Proof: Define the multiplicative group modulo p , i.e. $(\mathbb{Z}/p\mathbb{Z})^\times$. For all $a \in \mathbb{Z}$ (with operation being multiplication mod p), $\langle a \rangle \in (\mathbb{Z}/p\mathbb{Z})^\times$. Thus $|a| = |\langle a \rangle|$ must divide $p - 1$, so $a^{p-1} \equiv 1 \pmod{p}$.

Euler's Theorem: For all a relatively prime to n , $a^{\phi(n)} \equiv 1 \pmod{n}$, where ϕ is the Euler totient function.

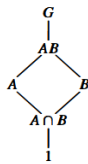
Proof: Define the multiplicative group mod n , $(\mathbb{Z}/n\mathbb{Z})^\times$. Also define the multiplicative group mod n of elements *relatively prime* to n , $(\mathbb{Z}/n\mathbb{Z})^*$. Then $|(\mathbb{Z}/n\mathbb{Z})^*| = \phi(n)$. Let a be relatively prime to n , so $a \in (\mathbb{Z}/n\mathbb{Z})^*$. Then all powers of a are also relatively prime to n , so $\langle a \rangle \in (\mathbb{Z}/n\mathbb{Z})^*$.

Use Lagrange's theorem to see that $|a| = |\langle a \rangle|$ must divide $\phi(n)$. Thus $a^{\phi(n)} \equiv 1 \pmod{n}$.

3.4 Isomorphism Theorems

1st Isomorphism Theorem: If $\phi : G \rightarrow H$ is a homomorphism, then $\ker \phi \trianglelefteq G$ and $G/\ker \phi \cong \phi(G)$.

2nd Isomorphism Theorem: Let $A, B \leq G$ and $A \leq N_G(B)$. Then $AB \leq G$, $B \trianglelefteq AB$, $A \cap B \trianglelefteq A$, and $AB/B \cong A/A \cap B$.



Lattice diagram of 2nd isomorphism theorem

3rd Isomorphism Theorem: Let $H, K \trianglelefteq G$ with $H \leq K$. Then $K/H \trianglelefteq G/H$ and

$$(G/H)/(K/H) \cong G/K$$

4th Isomorphism Theorem: Let $N \trianglelefteq G$. There's a bijection from the set of subgroups $A \leq G$ containing N onto the set of subgroups A/N of G/N .

I.e. every subgroup of G/N is of the form A/N for some $A \leq G$ containing N .

3.5 The Hölder Program

3.5.1 Composition Series

Call G a **simple group** if $|G| > 1$ and the only normal subgroups of G are 1 and G .

The sequence

$$1 = N_0 \leq N_1 \leq N_2 \leq \cdots \leq N_{k-1} \leq N_k = G$$

is called a **composition series** if $N_i \trianglelefteq N_{i+1}$ and N_{i+1}/N_i is a simple group. The quotient groups N_{i+1}/N_i are **composition factors** of G .

E.g. for the dihedral group D_8 ,

$$\begin{aligned} 1 \trianglelefteq \langle s \rangle \trianglelefteq \langle s, r^2 \rangle \trianglelefteq D_8 \\ 1 \trianglelefteq \langle r^2 \rangle \trianglelefteq \langle r \rangle \trianglelefteq D_8 \end{aligned}$$

3.5.2 Hölder Program

Jordan-Hölder Theorem: Let G be a finite group with $|G| > 1$. Then G has a composition series, and the composition factors are unique up to a permutation.

This suggests the **Hölder program**:

1. Classify all finite groups
2. Find all ways of combining two groups into one.

Theorem: There are 18 infinite simple group families and 26 simple groups not in those families (sporadic simple groups), such that every finite simple group is isomorphic to 1 in this list.

Feit-Thompson Theorem: If G is a simple group of odd order, then $G \cong Z_p$ for some prime p .

Proof: Both are trivial.

We call G **solvable** if it has the composition

$$1 = G_0 \trianglelefteq G_1 \trianglelefteq G_2 \trianglelefteq \cdots \trianglelefteq G_s = G$$

where G_{i+1}/G_i is Abelian.

3.6 The Alternating Group A_n

Call a 2-cycle $(i\ j)$ a **transposition**. Every element of S_n can be written as a product of transpositions.

Define the polynomial Δ and the permutation σ such that

$$\begin{aligned} \Delta &= \prod_{1 \leq i < j \leq n} (x_i - x_j) \\ \sigma(\Delta) &= \prod_{1 \leq i < j \leq n} (x_{\sigma(i)} - x_{\sigma(j)}) \end{aligned}$$

Since σ is a bijection, only a few terms $(x_{\sigma(i)} - x_{\sigma(j)})$ get their signs switched, so $\sigma(\Delta) = \pm\Delta$. Define ϵ such that

$$\epsilon(\sigma) = \begin{cases} +1 & \sigma(\Delta) = \Delta \\ -1 & \sigma(\Delta) = -\Delta \end{cases}$$

- E.g. for $\sigma = (1\ 2\ 3\ 4)$, we have $(1\ 2\ 3\ 4)(\Delta) = (-1)^3(\Delta)$, so $\epsilon((1\ 2\ 3\ 4)) = -1$. This is **not** the Levi-Civita symbol.
- $\epsilon(\sigma)$ is called the **sign** of σ .
- σ is an even permutation if $\epsilon(\sigma) = +1$, odd if $\epsilon(\sigma) = -1$.
- The map $\epsilon : S_n \rightarrow \{\pm 1\}$ is a homomorphism.

Theorem: transpositions $(i\ j)$ are all odd permutations.

Proof: Note that $\epsilon((1\ 2)) = -1$ since it only flips $(x_1 - x_2)$. For a general $(i\ j)$, let λ be the permutation exchanging 1 and i , 2 and j . Then $(i\ j) = \lambda(1\ 2)\lambda$. ϵ is a homomorphism, so

$$\epsilon((i\ j)) = \epsilon(\lambda(1\ 2)\lambda) = \epsilon(\lambda)^2\epsilon((1\ 2)) = -1$$

Define A_n , the **alternating group of degree n** , to be the kernel of ϵ . I.e. the set of even permutations. $S_n - A_n$ is the set of odd permutations.

By the 1st isomorphism theorem, $S_n/\ker \epsilon \cong \{\pm 1\}$, and by Lagrange's theorem, $|A_n| = n!/2$. A_n will become useful in the study of polynomials.

4 Group Actions

4.1 Definitions

Let G be a group acting on set A . For all $g \in G$, the map $\sigma_g : A \rightarrow A$ defined by $\sigma_g : a \mapsto g \cdot a$ is a permutation of A .

For each action of G on A , there's an associated homomorphism $\phi : G \rightarrow S_A$ defined by $\phi(g) = \sigma_g$, called the **permutation representation** associated with a given action.

- Kernel of action: $\{g \in G \mid g \cdot a = a, \forall a \in A\}$
- Stabilizer of a in G : $G_a = \{g \in G \mid g \cdot a = a\}$
- An action is **faithful** if the kernel is the identity.

A **permutation representation** of G is any homomorphism of G into the symmetric group S_A for some nonempty set A .

- Given any nonempty set A and homomorphism $\phi : G \rightarrow S_A$, the action of G on A can be $g \cdot a = \phi(g)(a)$. The kernel of this action is the same as the kernel of ϕ .

Let G act on set A . The relation on A defined by $a \sim b$ iff. $a = g \cdot b$ for some $g \in G$ is an **equivalence relation**. The **equivalence class** is $\mathcal{C}_a = \{g \cdot a \mid g \in G\}$. Alternatively, the **orbit of G containing a** .

- The orbit of a : set of all elements a can move to under a group action.
- $|\mathcal{C}_a| = |G : G_a|$, index of stabilizer a
- Action of G on A is **transitive** if there's only 1 orbit. I.e. for all $a, b \in A$, there exists $g \in G$ such that $a = g \cdot b$.

Cayley's Theorem: Every group is isomorphic to a subgroup of some symmetric group. If G has order n , then G is isomorphic to a subgroup of S_n .

Proof: Let G be a group, and let A be the set of left cosets of $1 \in G$. Let G act by left multiplication on A . The homomorphism $\pi : G \rightarrow S_G$ is that of a group action. The kernel of π is

$$\begin{aligned} \ker \pi &= \{g \in G \mid gx1 = x1 \quad \forall x \in G\} \\ &= \{1\} \end{aligned}$$

Thus π is an isomorphism.

4.2 Groups Acting on Themselves by Conjugation

4.2.1 Definitions

Let G act on $A = G$ through conjugation, such that

$$g \cdot a = gag^{-1}, \quad \forall g, a \in G$$

$a, b \in G$ are conjugate in G if there exists a $g \in G$ such that $b = g \cdot a = gag^{-1}$. I.e. if a and b are in the same orbit of G acting on itself by conjugation. These orbits are called the **conjugacy classes** of G .

- For an Abelian group, the conjugacy class of each $g \in G$ is $\{g\}$.
- Subsets $S, T \subseteq G$ are conjugate in G if there exists a $g \in G$ such that $T = gSg^{-1}$. I.e. if S, T are in the same orbit of G acting on itself by conjugation.

For $S \subseteq G$, if G acts on S with $g \cdot a = gag^{-1}, \forall a \in S$, the number of elements in the orbit/equivalence class containing S is $|G : G_S|$.

$$G_S = \{g \in G \mid gSg^{-1} = S\} = N_G(S)$$

- The number of conjugates of a subset $S \subseteq G$ is the index of the normalizer of S , i.e. $|G : N_G(S)|$.
- The number of conjugates of an element $s \in G$ is the index of the centralizer of s , i.e. $|G : C_G(s)|$.

The Class Equation: Let G be a finite group, and let $g_1, g_2, \dots, g_r$ be the representatives of distinct conjugacy classes of G not contained in the center $Z(G)$. Then

$$|G| = |Z(G)| + \sum_{i=1}^r |G : C_G(g_i)|$$

Corollary: If P is a group of prime power p^α for some $\alpha \geq 1$, then P has a nontrivial center $Z(P) \neq 1$.

Proof: By the class equation,

$$|P| = |Z(P)| + \sum_{i=1}^r |P : C_P(g_i)|$$

and by assumption, $C_P(g_i)$ is non-central, so $|C_P(g_i)| \neq |P|$. p divides $|P : C_P(g_i)|$, so p must divide $|Z(P)|$ and $|Z(P)| \neq 1$.

4.2.2 Conjugacy in Symmetric Groups

Let $\sigma, \tau \in S_n$ and let σ have the cycle decomposition $(a_1 \ a_2 \ \dots \ a_{k_1})(b_1 \ b_2 \ \dots \ b_{k_2}) \dots$. Then $\tau\sigma\tau^{-1}$ has the cycle decomposition

$$(\tau(a_1) \ \tau(a_2) \ \dots \ \tau(a_{k_1}))(\tau(b_1) \ \tau(b_2) \ \dots \ \tau(b_{k_2})) \dots$$

Proof: If $\sigma(i) = j$, then $\tau\sigma\tau^{-1}(\tau(i)) = \tau(j)$. Thus if i, j appears in the cycle decomposition of σ , then $\tau(i), \tau(j)$ appears in the cycle decomposition of τ .

If $\sigma \in S_n$ is the product of disjoint cycles of length $n_1, n_2, \dots, n_r$ with $n_1 \leq n_2 \leq \dots \leq n_r$, then $n_1, n_2, \dots, n_r$ are the **cycle type** of σ . Two elements of S_n are conjugate iff. they have the same cycle type.

The number of partitions of n equals the number of conjugacy classes of S_n . E.g. for $n = 5$,

Partition of 5	Representative of Conjugacy Class
1, 1, 1, 1, 1	1
1, 1, 1, 2	(1 2)
1, 1, 3	(1 2 3)
1, 4	(1 2 3 4)
5	(1 2 3 4 5)
1, 2, 2	(1 2)(3 4)
2, 3	(1 2)(3 4 5)

4.3 Automorphisms

An isomorphism of G onto itself is called an **automorphism**. The set of all automorphisms of G is $\text{Aut}(G)$.

Let $H \leq G$. Then G acts by conjugation on H as an automorphism of H . I.e. conjugation by $g \in G$ is an automorphism of H . The permutation representation of the homomorphism is $\phi : G \rightarrow \text{Aut}(H)$ with kernel $C_G(H)$.

$$h \mapsto ghg^{-1}, \quad \forall h \in H$$

Let $g \in G$. Conjugation by g is an **inner automorphism** of G , and the subgroup of $\text{Aut}(G)$ containing all inner automorphisms of G is called $\text{Inn}(G)$.

- Note that $\text{Inn}(G) \leq \text{Aut}(G)$, so $\text{Inn}(G) \cong G/Z(G)$.

A subgroup $H \leq G$ is **characteristic in** G , denoted $H \text{ char } G$, if every automorphism of G maps H to itself: $\sigma(H) = H$ for all $\sigma \in \text{Aut}(G)$.

- Characteristic subgroups are normal.
- Although normality isn't transitive, if $K \text{ char } H$ and $H \trianglelefteq G$, then $K \trianglelefteq G$.
- Characteristic subgroups are like “strongly normal” subgroups.

Theorem: For any $H \leq G$, $N_G(H)/C_G(H)$ is isomorphic to a subgroup of $\text{Aut}(H)$. If $H = G$, then this becomes that $G/Z(G)$ is isomorphic to a subgroup of $\text{Aut}(G)$.

Proof: Note that $H \trianglelefteq N_G(H)$. Conjugation by $g \in G$ is an automorphism of H with permutation representation $\phi : N_G(H) \rightarrow \text{Aut}(H)$ with kernel $C_G(H)$. By the 1st isomorphism theorem,

$$N_G(H)/C_G(H) \cong \text{Subgroup of } \text{Aut}(H)$$

Theorem: The automorphism group of cycle groups of order n , Z_n , is isomorphic to $(\mathbb{Z}/n\mathbb{Z})^\times$, an Abelian group of order $\phi(n)$.

Proof: Let $Z_n = \langle x \rangle$. If $\psi \in \text{Aut}(Z_n)$, then $\psi(x) = x^a$ for some $a \in \mathbb{Z}$. Further, $|x| = n$ so a is defined mod n . We seek to show a bijection between Z_n and $(\mathbb{Z}/n\mathbb{Z})^\times$. Define the homomorphism (it can be shown that $\Psi(\psi_a \circ \psi_b) = \Psi(\psi_a) \circ \Psi(\psi_b)$)

$$\begin{aligned} \Psi : \text{Aut}(Z_n) &\rightarrow (\mathbb{Z}/n\mathbb{Z})^\times \\ \psi_a &\mapsto a \pmod{n}, \quad \psi_a \in \Psi \end{aligned}$$

This map is surjective. It is also injective, thus Ψ is an isomorphism.

4.4 Sylow Theorems

Let p be a prime. A group of order p^α for some $\alpha \geq 1$ is called a p -group. A subgroup of G that is a p -group is called a p -subgroup.

If $|G| = p^\alpha m$ where $p \nmid m$. A subgroup of order p^α (i.e. max subgroup of size that's a power of p) is called a **Sylow p -subgroup of G** .

- Set of Sylow p -subgroups of G : $\text{Syl}_p(G)$
- Number of Sylow p -subgroups of G : $n_p(G)$

Sylow Theorems: (partial converse to Lagrange's Theorem) Let $|G| = p^\alpha m$, where p is prime and $p \nmid m$. Then

1. $\text{Syl}_p(G) \neq \emptyset$
2. If p is a Sylow p -subgroup of G , and Q is any p -subgroup of G , then there exists $g \in G$ such that $Q \leq gPg^{-1}$. Further, any 2 Sylow p -subgroups of G are conjugate in G .
3. The number of Sylow p -subgroups of G is of the form $1 + kp$, i.e. $n_p \equiv 1 \pmod{p}$. Further, $n_p = |G : N_G(P)|$ for a Sylow p -subgroup P . Thus n_p divides m .

If P is a Sylow p -subgroup of G , the following are equivalent:

- 1) P is the unique Sylow p -subgroup of G , $n_p = 1$
- 2) $P \trianglelefteq G$
- 3) $P \text{ char } G$
- 4) All subgroups generated by elements of power p^n for some n are p -groups.

4.4.1 Applications

Theorem: If $|G| = pq$ for prime p, q with $p < q$, let $P \in \text{Syl}_p(G)$ and $Q \in \text{Syl}_q(G)$. Then $Q \trianglelefteq G$. If $P \trianglelefteq G$ as well, then G is cyclic.

Proof: (1) Since $Q \in \text{Syl}_q(G)$, $n_q = 1 + kq$ for some $k \geq 0$. And n_q divides p , so $p < q$ and $p = \text{prime}$ enforces $k = 0$. Thus $n_q = 1$, and this is equivalent to $Q \trianglelefteq G$.

Now since $P \in \text{Syl}_p(G)$, $n_p = 1 + kp$ for some $k \geq 0$. Because n_p divides q , either $n_p = 1$ or $n_p = q$. If $n_p = q$, then $q \equiv 1 \pmod{p}$. If $n_p = 1$, then q is not $1 \pmod{p}$, and $P \trianglelefteq G$.

- (2) If $P \trianglelefteq G$, then $n_p = 1$ so q is not $1 \pmod{p}$. Since P, Q have prime power, they're cyclic: $P = \langle x \rangle$ and $Q = \langle y \rangle$. If $P \trianglelefteq G$, $N_G(P) = G$, so $G/C_G(P) \cong \text{Subgroup of } \text{Aut}(P) = \text{Subgroup of } Z_p$. This has order $(p - 1)$, so by Lagrange's theorem,

$$\frac{|G|}{|C_G(P)|} = |Z_p| = p - 1$$

Neither p, q divide $p - 1$, so $G = C_G(P)$. Thus for $x \in P \leq Z_G(P)$, x commutes with all y , and $|xy| = pq$ or $G \cong Z_{pq}$.

E.g. Groups of Order 30: We show that all groups of order 30 have a normal subgroup isomorphic to Z_{15} .

Sketch: Any subgroup of order 15 is normal, since it has index 2. Also cyclic by previous result, since $15 = 3 \cdot 5$ and $3 \nmid (5 - 1)$. So we just show that there exists a subgroup of order 15.

Let $P \in \text{Syl}_p(G)$ and $Q \in \text{Syl}_q(G)$. Assume neither are normal. The possibilities then are $n_5 = 6$, $n_3 = 10$, not $n_5 = n_3 = 1$. There are $4 \cdot 6 = 24$ non-identity elements of order 5, and $2 \cdot 10$ non-identity elements of order 3. Total 44, yet $|G| = 30$, impossible. Thus at least 1 of P, Q must be normal, and

$$|PQ| = \frac{|P||Q|}{|P \cap Q|} = 15$$

5 Direct and Semidirect Products and Abelian Groups

5.1 Direct Products

The **direct product** $G_1 \times G_2 \times \cdots \times G_n$ of groups G_i , $1 \leq i \leq n$, with operations $\star_1, \star_2, \dots, \star_n$, is the set of n -tuples $(g_1, g_2, \dots, g_n)$ where $g_i \in G_i$ and the operation is defined componentwise:

$$(g_1, g_2, \dots, g_n) \star (h_1, h_2, \dots, h_n) = (g_1 \star_1 h_1, g_2 \star_2 h_2, \dots, g_n \star_n h_n)$$

And likewise for infinitely many n . For example, if $G_i = (\mathbb{R}, +)$, then $\mathbb{R} \times \mathbb{R} \times \cdots \times \mathbb{R}$ is the Euclidean n -space $\mathbb{R}^n$ such that

$$(a_1, a_2, \dots, a_n)(b_1, b_2, \dots, b_n) = (a_1 + b_1, a_2 + b_2, \dots, a_n + b_n)$$

- If $G_1, G_2, \dots, G_n$ are groups, their direct product is a group of order $|G_1||G_2| \cdots |G_n|$.
- The order of an element $x_1 x_2 \dots x_n \in G$ is $\text{lcm}(|x_1|, |x_2|, \dots, |x_n|)$.
- For a fixed i , the set of elements of G with 1_{G_j} in the j -th position for all $i \neq j$ is a subgroup of G isomorphic to G_i .

$$G_i \cong \{(1, 1, \dots, 1, g_i, 1, \dots, 1) \mid g_i \in G_i\}$$

5.2 Fundamental Theorem of Finitely-Generated Abelian Groups

G is **finitely generated** if there's a finite subset A of G such that $G = \langle A \rangle$. For each $r \in \mathbb{Z}$ with $r \geq 0$, let $\mathbb{Z}^r = \mathbb{Z} \times \cdots \times \mathbb{Z}$ be the direct product of r $\mathbb{Z}$'s. Then $\mathbb{Z}^r$ is the **free Abelian group of rank r** .

Fundamental Theorem of Finitely Generated Abelian Groups: Let G be a finitely generated Abelian group. Then

$$G \cong \mathbb{Z}^r \times Z_{n_1} \times Z_{n_2} \times \cdots \times Z_{n_s}$$

for integers $r, n_1, n_2, \dots, n_s$ satisfying

1. $r \geq 0$ and $n_i \geq 2$ for all i .
2. n_{i+1} divides n_i for all applicable i .

The decomposition of G in this way is unique, called the **invariant factor decomposition of G** . r is the **free rank/Betti number** of G . Integers $n_1, n_2, \dots, n_s$ are the **invariant factors** of G .

The order of a finite Abelian group is the product of its invariant factors. G is said to be of type $(n_1, n_2, \dots, n_s)$, and s is the **rank of G** .

E.g. For an Abelian group of order 180, the possible invariant factors are

Invariant Factors	Abelian Groups
$2^2 \cdot 3^2 \cdot 5$	Z_{180}
$2 \cdot 3^2 \cdot 5, 2$	$Z_{90} \times Z_2$
$2^2 \cdot 3 \cdot 5, 3$	$Z_{60} \times Z_3$
$2 \cdot 3 \cdot 5, 2 \cdot 3$	$Z_{30} \times Z_6$

Theorem: Let G be an Abelian group of order $n > 1$ with unique prime factorization

$$n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k}$$

Then $G \cong A_1 \times A_2 \times \cdots \times A_k$, with $|A_i| = p_i^{\alpha_i}$. For each $A \in \{A_1, \dots, A_k\}$ with $|A| = p^\alpha$,

$$A \cong Z_{p^{\beta_1}} \times Z_{p^{\beta_2}} \times \cdots \times Z_{p^{\beta_t}}$$

where $\beta_1 \geq \beta_2 \geq \dots \geq \beta_t \geq 1$ and $\beta_1 + \beta_2 + \dots + \beta_t = \alpha$. **This decomposition is unique.** p^{β_i} are the **elementary divisors** of G , and the decomposition is the **elementary divisor decomposition** of G .
I.e. G is the direct product of its Sylow p -subgroups.

E.g. For the Abelian group of order $n = 1800 = 2^3 \cdot 3^2 \cdot 5^2$,

Order p^β	Partitions of β	Abelian Groups
2^3	$(3), (2, 1), (1, 1, 1)$	$Z_8, Z_4 \times Z_2, Z_2 \times Z_2 \times Z_2$
3^2	$(2), (1, 1)$	$Z_9, Z_3 \times Z_3$
5^2	$(2), (1, 1)$	$Z_{25}, Z_5 \times Z_5$

Choose 1 Abelian group from each row. E.g. $G \cong Z_8 \times Z_3 \times Z_3 \times Z_{25}$. This **elementary divisor decomposition** can convert to **invariant factor decomposition** and vice versa.

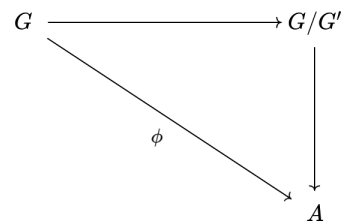
5.3 Recognizing Direct Products

Let $x, y \in G$ and $A, B \subseteq G$ be nonempty subsets. Then $[x, y] = x^{-1}y^{-1}xy$ is called the **commutator** of x and y .

- $[A, B] = \langle [a, b] \mid a \in A, b \in B \rangle$ is the group generated by commutation of elements from A and B .
- $G' = \langle [x, y] \mid x, y \in G \rangle$ is the **commutation subgroup** of G .

Let $x, y \in G$ and $H \leq G$. Then

- $H \trianglelefteq G$ iff. $[H, G'] \leq H$
- $\sigma[x, y] = [\sigma(x), \sigma(y)]$ for any automorphism σ of G .
- If $\phi : G \rightarrow A$ is the homomorphism of G into Abelian group A , then ϕ factors through G' ,



i.e. $G' \leq \ker \phi$. The following diagram commutes: |
 diagram commutes |

| |:-| | This

Recognition Theorem: Let H, K be subgroups of G such that $H, K \trianglelefteq G$ and $H \cap K = 1$. Then

$$HK \cong H \times K$$

The LHS is the **internal direct product** with elements hk . The RHS is the **external direct product** with elements (h, k) .

Part II

Ring Theory

6 Introduction to Rings

6.1 Definitions

A ring is a set R with binary operations $+$, $\times$ satisfying

1. $(R, +)$ is an Abelian group
2. $\times$ is associative, $(a \times b) \times c = a \times (b \times c)$ for all $a, b, c \in R$
3. Distributive law holds:

$$(a + b) \times c = (a \times c) + (b \times c)$$

$$a \times (b + c) = (a \times b) + (a \times c)$$

- If $\times$ is commutative, R is called a **commutative ring**.
- R has identity 1 if there exists $1 \in R$ such that $1 \times a = a \times 1 = a$ for all $a \in R$.
- A subgroup of R closed under multiplication is called a **subring of R** . E.g. $\mathbb{Z}$ is a subring of $\mathbb{Q}$, and $\mathbb{Q}$ is a subring of $\mathbb{R}$.

Let R be a ring. A nonzero $a \in R$ is a **zero divisor** if there exists $b \in R$ such that $ab = 0$ or $ba = 0$.

Let R have identity 1. An element $u \in R$ is a **unit in R** if there exists $v \in R$ such that $uv = vu = 1$. The set of units in R is $R^\times$. Note that zero divisors cannot be a unit.

A ring R with identity $1 \neq 0$ is called a **division ring/skew field** if every nonzero $a \in R$ has a multiplicative inverse. A commutative division ring is called a **field**.

- Or: a field is a commutative ring with identity $1 \neq 0$ in which every nonzero element is a unit.
 - I.e. a field contains no zero divisors. $\mathbb{R}$ is an example of a field, but $\mathbb{Z}$ is not.
-

A commutative ring with identity $1 \neq 0$ and no zero divisors is called an **integral domain**. I.e. rings with same structure as $\mathbb{Z}$.

Theorem: Any finite integral domain is a field.

Proof: Let R be a finite integral domain, with nonzero $a \in R$. Then $x \mapsto ax$ is an injection. Since R is finite, $x \mapsto ax$ is also surjective, and thus there exists $b \in R$ such that $ab = 1$. I.e. a is a unit in R , and since a was arbitrary, R is a field.

6.2 Types of Rings

6.2.1 Polynomial Rings

Let R be a commutative ring with identity. The sum $a_n x^n + \cdots + a_1 x + a_0$ with $a_n \neq 0$ and $n \geq 0$ is the polynomial in x with coefficients $a_i \in R$.

The set of all such polynomials is called the **ring of polynomials** in the variable x with coefficients in R , denoted $R[x]$.

Note that $R \subset R[x]$. R is the subset of constant polynomials.

6.2.2 Matrix Rings

Fix a ring R (doesn't need to be commutative) and let $n \in \mathbb{Z}^+$. Let $M_n(R)$ be the set of all $n \times n$ matrices with entries from R . This is the **matrix ring**.

- If R is nontrivial and $n \geq 2$, then $M_n(R)$ is not commutative.
- Even if R has no zero divisors, $M_n(R)$ with $n \geq 2$ can have zero divisors. Easy to construct.

6.2.3 Group Rings

Fix a commutative ring R with identity $1 \neq 0$ and let $G = \{g_1, g_2, \dots, g_n\}$ be a finite group. **Group rings** RG of G with coefficient in R is the set of sums

$$a_1 g_1 + a_2 g_2 + \cdots + a_n g_n, \quad a_i \in R, \quad 1 \leq i \leq n$$

E.g. for the dihedral group D_8 , the group ring $\mathbb{Z}D_8$ has elements like $r + r^2 - 2s, -3r^2 + rs$, etc.

If $|G| > 1$, then RG always has zero divisors. E.g. let $g \in G$ and $|g| = m$. Then

$$(1 - g)(1 + g + \cdots + g^{m-1}) = 1 - g^m = 0$$

If S is a subring of R , then SG is a subring of RG . Similarly, if H is a subgroup of G , then RH is a subring of RG .

6.3 Ring Homomorphisms

A ring homomorphism is a map from 1 ring to another, respecting the additive and multiplicative structures. Let R, S be rings. $\phi : R \rightarrow S$ is a ring homomorphism if

$$\begin{aligned}\phi(a + b) &= \phi(a) + \phi(b) \\ \phi(ab) &= \phi(a)\phi(b)\end{aligned}$$

The kernel of ϕ is denoted $\ker \phi$, the set of elements of R that map to $0 \in S$.

The homomorphism is called an isomorphism if it's bijective.

- E.g. $\phi : \mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z}$ is a ring homomorphism (respects additive and multiplicative structures).
- E.g. $\phi : \mathbb{Q}[x] \rightarrow \mathbb{Q}$ with $\phi(p(x)) = p(0)$ for $p(x) \in \mathbb{Q}[x]$ is a ring homomorphism.

6.3.1 Ring Ideals

Previously, normal subgroups were the kernels of group homomorphisms. Now, ideals will be kernels of ring homomorphisms.

Let I be an arbitrary subset of the ring R , and $r \in R$. I is the **left ideal** of R if

1. I is a subring of R (closed under $+$, $\times$)
2. I is closed under left multiplication, $rI \subseteq I$ for all $r \in R$.

Right ideals are defined similarly. If I is both a left and right ideal, it's a **two-sided ideal** of R .

Ideals are kernels of ring homomorphisms.

- Fibers of ϕ : $\{r + I \mid r \in R\}$
- The fibers of ϕ have a structure **isomorphic** to the image of ϕ .
- I.e. if X is fiber over $a \in R$ and Y is fiber over $b \in R$, then $X + Y$ is fiber over $a + b$ and XY is fiber over ab .

The ring of cosets, R/I for an ideal I , is called the **quotient ring** of R by $I = \ker \phi$. R/I is a ring under (for $r, s \in R$):

$$\begin{aligned}(r + I) + (s + I) &= (r + s) + I \quad \text{or} \quad \bar{r} + \bar{s} = \overline{r + s} \\ (r + I) \times (s + I) &= (rs) + I \quad \text{or} \quad \bar{r}\bar{s} = \overline{rs}\end{aligned}$$

6.3.2 The Isomorphism Theorems

1st Isomorphism Theorem: If $\phi : R \rightarrow S$ is a homomorphism of rings, then:

1. $\ker \phi$ is an ideal of R .
2. Image of the homomorphism, $\phi(R)$, is a subring of S .
3. $R/\ker \phi \cong \phi(R)$

2nd Isomorphism Theorem: Let A be a subring and B an ideal of R . Then:

1. $A + B$ is a subring of R .
2. $A \cap B$ is an ideal of A .
3. $(A + B)/B \cong A/(A \cap B)$

3rd Isomorphism Theorem: Let I, J be an ideal of R with $I \subseteq J$. Then J/I is an ideal of R/I and

$$(R/I)/(J/I) \cong R/J$$

4th Isomorphism Theorem: Let I be an ideal of R . Define a correspondence $A \leftrightarrow A/I$ between the set of subrings A of R that contain I , and the set of subrings of R/I . *This is a bijection.* A (subring containing I) is an ideal of R iff. A/I is the ideal of R/I .

6.3.3 Solving Diophantine Equations with $\phi : \mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$

Note that $\phi : \mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$ is a ring homomorphism formed by factoring out the ideal $n\mathbb{Z}$. We can use this to solve Diophantine equations.

E.g. $x^2 + y^2 = 3z^2$. This is a relation between elements of the ring $\mathbb{Z}$, thus the relation must also hold in any quotient ring since the structure is preserved. We assume $\gcd(x, y, z) = 1$ since otherwise we can get a more reduced equation.

Consider $\phi : \mathbb{Z} \rightarrow \mathbb{Z}/4\mathbb{Z}$, since the squares $0^2, 1^2, 2^2, 3^2$ are just $0, 1 \pmod{4}$. Thus in $\mathbb{Z}/4\mathbb{Z}$, the equation is

$$\begin{Bmatrix} 0 \\ 1 \end{Bmatrix} + \begin{Bmatrix} 0 \\ 1 \end{Bmatrix} = 3 \begin{Bmatrix} 0 \\ 1 \end{Bmatrix} \equiv \begin{Bmatrix} 0 \\ 3 \end{Bmatrix} \pmod{4}$$

The only solution is $x^2, y^2, z^2 = 0 \pmod{4}$, so by checking choices, they must all be even (0 or 2 mod 4). But then $\gcd(x, y, z) \neq 1$, a contradiction, thus this equation has no solution in $\mathbb{Z}$.

6.4 Properties of Ideals

6.4.1 Finitely Generated Ideals

Let A be a subset of a ring R . (A) denotes the smallest ideal of R containing A and is called the **ideal generated by A** .

- Principal ideal: ideal generated by 1 element
- Finitely generated ideal: ideal generated by finite set

$$(A) = \bigcap_{I \text{ an ideal}, A \subseteq I} I$$

E.g. $\{0\}$ and R are principal ideals: $0 = (0)$ and $R = (1)$.

E.g. $n\mathbb{Z}$ is a principal idea: $n\mathbb{Z} = (n)$.

6.4.2 Maximal Ideals

Theorem: Let I be an ideal of R .

- 1) $I = R$ iff. I contains a unit.
- 2) If R is commutative, R is a field iff its only ideals are 0 and R .

Proof: (1) If $I = R$, I contains unit 1. If u is a unit in I with inverse v , for any $r \in R$,

$$r = r \cdot 1 = r(vu) = (rv)u \in I$$

- (2) If R is a field, every nonzero element must be a unit, i.e. every nonzero ideal contains a unit. By part (1), R is the only nonzero ideal. Other direction: if $0, R$ are only ideals of R , let $u \in R$, $u \neq 0$. R is commutative, so $R = (u)$ and $1 \in (u)$, i.e. $1 = vu$, i.e. u is a unit. u is arbitrary, so R is a field.

An ideal M in arbitrary ring R is called a **maximal ideal** if $M \neq R$ and the only ideals containing M are M and R .

- In a ring with identity, every proper ideal is contained in a maximal ideal.
- If R is commutative, the ideal M is maximal iff. R/M is a field. (proof: 4th isomorphism theorem + previous theorem)

Above gives a procedure for constructing fields: 1) Get a ring, 2) Divide by maximal ideal. E.g. $n\mathbb{Z}$ is a maximal ideal of $\mathbb{Z}$ iff. $\mathbb{Z}/n\mathbb{Z}$ is a field, i.e. n is prime.

6.4.3 Prime Ideals

Let R be a commutative ring. Ideal P is a **prime ideal** if $P \neq R$ and whenever $ab \in P$ for $a, b \in R$, at least one of a, b is in P .

E.g. ideal $n\mathbb{Z}$ with $n > 1$ is a prime ideal of $\mathbb{Z}$ iff. for every time $ab \in n\mathbb{Z}$ for $a, b \in \mathbb{Z}$, at least 1 of a, b is in $n\mathbb{Z}$. I.e. whenever n divides ab , n divides either a or b or both. I.e. n is prime — and $p\mathbb{Z}$ are the prime ideals of $\mathbb{Z}$.

0 can also be a prime ideal.

Theorem: If R is commutative, the ideal P is a prime ideal in R iff. R/P is an integral domain.

Proof: P is prime iff. whenever $ab \in P$, either $a \in P$ or $b \in P$. Let $\bar{r} = r + P$. Then $r \in P$ iff. $\bar{r} = 0$ in R/P . Thus P is a prime ideal iff. whenever $\bar{a}\bar{b} = \bar{a}\bar{b} = \bar{0}$, $\bar{a} = 0$ or $\bar{b} = 0$. I.e. R/P is an integral domain, no zero divisors.

Theorem: If R is commutative, every maximal ideal of R is a prime ideal.

Proof: If M is a maximal ideal, R/M is a field by previous theorems. Every field is an integral domain, thus M is a prime ideal.

- E.g. $p\mathbb{Z}$ is both a prime and maximal ideal in $\mathbb{Z}$.
- Note that above doesn't work in reverse. E.g. (x) is a prime ideal in $\mathbb{Z}[x]$ since $\mathbb{Z}[x]/(x) \cong \mathbb{Z}$, an integral domain. But this is not a field, so (x) is not a maximal ideal in $\mathbb{Z}[x]$.
- The problem is that not all “integral domains” are “fields” (but field = finite integral domain works).

6.5 Rings of Fractions

Let R be a commutative ring. Let D be a nonempty subset of R that doesn't contain zero or any zero divisors, and is closed under multiplication. Then there exists a commutative ring Q with $1 \in Q$ such that R is a subring of Q and every element of D is a unit in Q .

- Every element of Q is of the form rd^{-1} for $r \in R$ and $d \in D$.
- If $D = R - \{0\}$, then Q is a field.
- Q is unique up to an isomorphism.

Q is called a “**ring of fractions**” of D w.r.t R . Denoted $Q = D^{-1}R$. Essentially, the smallest field that can be built by taking ratios of integral domains.

If R is an integral domain and $D = R - \{0\}$, then Q is a **field of fractions** or **quotient field** of R .

- E.g. if R is a field, its field of fractions is just R .
- The field of fractions of $\mathbb{Z}$ is $\mathbb{Q}$.
- The field of fractions of $2\mathbb{Z} \subset \mathbb{Z}$ is $\mathbb{Q}$.

6.6 Chinese Remainder Theorem

A ring direct product is $R_1 \times R_2$ with ordered pairs (r_1, r_2) such that

$$\begin{aligned}(r_1, r_2) + (s_1, s_2) &= (r_1 + s_1, r_2 + s_2) \\ (r_1, r_2)(s_1, s_2) &= (r_1 s_1, r_2 s_2)\end{aligned}$$

Ideals A, B of R are called **comaximal** if $A + B = R$. Then $A \cap B = AB$.

Chinese Remainder Theorem: Let $A_1, A_2, \dots, A_k$ be ideals in R . The map

$$R \rightarrow R/A_1 \times R/A_2 \times \dots \times R/A_k$$

defined by $r \mapsto (r + A_1, r + A_2, \dots, r + A_k)$ is a ring homomorphism with kernel $A_1 \cap A_2 \cap \dots \cap A_k$. If A_i, A_j are comaximal for every $i \neq j$, then $A_1 \cap A_2 \cap \dots \cap A_k = A_1 A_2 \dots A_k$. Thus

$$R/(A_1 A_2 \dots A_k) \cong R/A_1 \times R/A_2 \times \dots \times R/A_k$$

The special case is $\mathbb{Z}/mn\mathbb{Z} \cong (\mathbb{Z}/m\mathbb{Z}) \times (\mathbb{Z}/n\mathbb{Z})$ for coprime m, n .

Let $n = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_k^{\alpha_k}$ for prime p_i . Then

$$(\mathbb{Z}/n\mathbb{Z}) \cong (\mathbb{Z}/p_1^{\alpha_1}\mathbb{Z}) \times (\mathbb{Z}/p_2^{\alpha_2}\mathbb{Z}) \times \dots \times (\mathbb{Z}/p_k^{\alpha_k}\mathbb{Z})$$

This is an isomorphism of rings, so a group of units should also obey the isomorphism. Thus

$$\begin{aligned}(\mathbb{Z}/n\mathbb{Z})^\times &\cong (\mathbb{Z}/p_1^{\alpha_1}\mathbb{Z})^\times \times (\mathbb{Z}/p_2^{\alpha_2}\mathbb{Z})^\times \times \dots \times (\mathbb{Z}/p_k^{\alpha_k}\mathbb{Z})^\times \\ \phi(n) &= \phi(p_1^{\alpha_1})\phi(p_2^{\alpha_2}) \dots \phi(p_k^{\alpha_k})\end{aligned}$$

7 Domains, Domains, Domains

7.1 Euclidean Domains

7.1.1 Definitions

We define a norm as a measure of “size” in rings. A **norm** is any function $N : R \rightarrow \mathbb{Z}^+ \cup \{0\}$ with $N(0) = 0$. If $N(a) > 0$ for all $a \neq 0$, then N is a **positive norm**. This definition is very loose.

An integral domain R is a **Euclidean domain** if there exists a norm N on R such that for any $a, b \in R$ with $b \neq 0$, there exists $q, r \in R$ such that

$$\begin{aligned} a &= qb + r, \quad r = 0 \text{ or } N(r) < N(b) \\ q &= \text{quotient} \\ r &= \text{remainder} \end{aligned}$$

I.e. R is a Euclidean domain if there exists a division algorithm.

- $\mathbb{Z}$ is a Euclidean domain with norm $N(a) = |a|$.
- If F is a field, $F[x]$ is a Euclidean domain with norm $N(p(x)) = \text{degree of } p(x)$.

Theorem: Every ideal of a Euclidean domain is principal. I.e. if I is any nonzero ideal in a Euclidean domain R , then $I = (d)$ where d is a nonzero, minimum-norm element of I .

Proof: Let d be a nonzero element of I with minimum norm, which exists by well-ordering of $\mathbb{Z}$. Then $(d) \subseteq I$. Conversely, let $a \in I$ and use the division algorithm to write $a = qd + r$ with $N(r) < N(d)$. Then $r = a - qd$. Both a, qd are in I by definition, so $r \in I$. Since $N(r) < N(d)$ and $N(d)$ is minimal, $r = 0$. Thus $a = qd \in (d)$, i.e. $I \subseteq (d)$.

- E.g. Every ideal $n\mathbb{Z}$ of the Euclidean domain $\mathbb{Z}$ are principal. $n\mathbb{Z} = (n)$.
- The ring of polynomials $\mathbb{Q}[x]$ is a Euclidean domain since $\mathbb{Q}$ is a field.
- But the ring of polynomials $\mathbb{Z}[x]$ is not a Euclidean domain since $\mathbb{Z}$ is not a field. Or: $(2, x)$ is an ideal of $\mathbb{Z}[x]$ that is not principal, hence by above, $\mathbb{Z}[x]$ cannot be a Euclidean domain.

7.1.2 Greatest Common Divisors

A Euclidean domain provides the notion of a GCD. If I is an ideal generated by a, b and $I = (d)$ where the norm of d is maximal, then d is the GCD of a, b .

The GCD can be computed algorithmically. Let R be the Euclidean domain with $a, b \in R$ and $a, b \neq 0$. Following the **Euclidean algorithm**,

$$\begin{aligned} a &= q_0b + r_0 \\ b &= q_1r_0 + r_1 \\ r_0 &= q_2r_1 + r_2 \\ &\vdots \\ r_{n-2} &= q_nr_{n-1} + r_n \\ r_{n-1} &= q_{n+1}r_n \end{aligned}$$

Then $\gcd(a, b) = r_n$. This is a very fast algorithm.

7.2 Principal Ideal Domains

A **principal ideal domain** is an integral domain in which every ideal is principal.

Theorem: Every nonzero prime ideal in a PID is a maximal ideal.

Proof: Let (p) be a nonzero prime ideal in R , and let $I = (m)$ be an ideal containing (p) . We seek to show that either $I = (p)$ or $I = R$. Since $p \in (m)$, $p = rm$ for some $r \in R$. (p) is a prime ideal, so either $r \in (p)$ or $m \in (p)$.

- If $r \in (p)$, then $r = ps$, $p = rm = psm$ and $sm = 1$. m is a unit, thus $I = R$.
- If $m \in (p)$, then $(m) = (p) = I$.

Theorem: If R is a commutative ring such that $R[x]$ is a PID, then R is a field.

Proof: If $R[x]$ is a PID, then R is an integral domain. Since $R[x]/(x) \cong R$, (x) is a nonzero prime ideal in $R[x]$. By the previous theorem, (x) is a maximal ideal, thus R is a field.

7.2.1 Dedekind-Hasse Norm

N is a **Dedekind-Hasse norm** if

- 1) N is a positive norm
- 2) For all $a, b \in R$, either $a \in (b)$ or $\exists s, t \in R$ such that $0 < N(sa - tb) < N(b)$.

This is a weak version of the Euclidean condition for Euclidean domains.

An integral domain is a PID iff. it has a Dedekind-Hasse norm.

- Thus not every PID is an ED (Dedekind-Hasse norm $\neq$ Euclidean norm)
- But every ED is a PID (Euclidean norms satisfy Dedekind-Hasse norm, e.g. $s = 1$)

7.3 Unique Factorization Domains

7.3.1 Definitions

Let R be an integral domain. Let $r \in R$ with r not a unit.

1. r is **irreducible** in R if whenever $r = ab$ for $a, b \in R$, at least 1 of a, b is a unit.
2. $p \in R$ is a **prime** in R if the ideal (p) is a prime ideal. I.e. $p \neq 0$ is prime if it's not a unit, and whenever p divides ab , it divides at least a or b .
3. $a, b \in R$ differing by a unit ($a = ub$) are **associates** in R .

*In an integral domain, a prime element is irreducible.

*In a PID, a nonzero element is prime iff. it's irreducible.

In $\mathbb{Z}$, irreducible elements are primes and their negatives. a, b are associates if $a = \pm b$.

Unique Factorization Domain (UFD): An integral domain in which every nonzero $r \in R$ obeys:

1. r can be written as a finite product of irreducibles $p_i \in R$.

$$r = p_1 p_2 \cdots p_n$$

2. This decomposition is unique up to associates.

Fundamental Theorem of Arithmetic: $\mathbb{Z}$ is a UFD.

- In a UFD, prime = irreducible.
- A field F is a UFD since all elements are units.
- E.g. $\mathbb{Z}[\sqrt{-5}]$ is not a UFD, since $6 = 2 \cdot 3 = (1 + \sqrt{-5})(1 - \sqrt{-5})$ and the decomposition is not unique.

Our domains so far:

Fields $\subset$ Euclidean Domain $\subset$ PID $\subset$ UFD $\subset$ Integral Domain

7.3.2 Fermat's Theorem on Sums of Squares

(A sketch of the application of ring theory to number theory.)

Fermat's Theorem on Sums of Squares: The prime p satisfies $p = a^2 + b^2$ iff. $p = 2$ or $p \equiv 1 \pmod{4}$.

Define the **quadratic integer ring** $\mathcal{O}$ with squarefree integer D such that

$$\mathcal{O}_{\mathbb{Q}(\sqrt{D})} = \mathbb{Z}[\omega] = \{a + b\omega \mid a, b \in \mathbb{Z}\}$$

$$\omega = \begin{cases} \sqrt{D} & D \equiv 2, 3 \pmod{4} \\ \frac{1+\sqrt{D}}{2} & D \equiv 1 \pmod{4} \end{cases}$$

If $D = -1$, we get the **ring of Gaussian integers** $\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}$. The field norm $N : \mathbb{Q}(\sqrt{D}) \rightarrow \mathbb{Q}$ is defined as

$$N(a + b\sqrt{D}) = (a + b\sqrt{D})(a - b\sqrt{D}) = a^2 - Db^2 \in \mathbb{Q}$$

If $D = -1$, the norm is just $a^2 + b^2$.

Lemma: If $N(\alpha)$ is $\pm$ prime in $\mathbb{Z}$, then α is irreducible in $\mathcal{O}$.

Proof: If $\alpha = \beta\gamma$ for $\beta, \gamma \in \mathcal{O}$, then $N(\alpha) = N(\beta)N(\gamma)$ is prime. One of $N(\beta)$, $N(\gamma)$ must be ± 1 and the other is $\pm p$. An element of $\mathcal{O}$ has norm ± 1 iff. it's a unit in $\mathcal{O}$, so α is irreducible.

For the $D = -1$ case of $\mathbb{Z}[i]$: Since $\mathbb{Z}[i]$ is a Euclidean domain, PID, and UFD, an irreducible element is prime. $\alpha = a + bi$ has norm $\alpha\bar{\alpha} = a^2 + b^2$. Thus α is irreducible in $\mathbb{Z}[i]$ iff. $p = a^2 + b^2$ is prime in $\mathbb{Z}$.

$p = 2$ obviously works: $2 = 1^2 + 1^2$. A square of an integer is always 0 or 1 mod 4, so p (a prime) must be 1 mod 4. For exact details see Dummit & Foote 8.3.

8 Polynomial Rings

8.1 Definitions

The **polynomial ring** $R[x]$ with coefficients from ring R is the set of elements $a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0$ with $a_i \in R$. $R[x]$ is an integral domain.

Theorem: Let I be an ideal of the ring R . Let (I) be the ideal of $R[x]$ generated by I . Then

$$R[x]/(I) \cong (R/I)[x]$$

Proof: The transformation $\phi : R[x] \rightarrow (R/I)[x]$ is a homomorphism by reducing each coefficient mod I . The kernel of ϕ is $\ker \phi = (I)$, and the result follows from the isomorphism theorems.

- E.g. for an ideal $n\mathbb{Z}$ of $\mathbb{Z}$, we have $\mathbb{Z}[x]/n\mathbb{Z}[x] \cong (\mathbb{Z}/n\mathbb{Z})[x]$.
-

We can define a polynomial ring in several variables inductively:

$$R[x_1, x_2, \dots, x_n] = R[x_1, x_2, \dots, x_{n-1}][x_n]$$

- Terms have the form $ax_1^{d_1}x_2^{d_2}\cdots x_n^{d_n}$ with $a \in R$.
- d_i is the degree in x_i .
- $d = d_1 + d_2 + \cdots + d_n$ is the degree of the term.
- $(d_1, d_2, \dots, d_n)$ is the multidegree of the term.

E.g. $\mathbb{Z}[x, y]$ has elements $p(x, y) = 2x^3 + 67xy$ of degree 3 and $q(x, y) = 163x^2y^3$ of degree 5.

8.1.1 Polynomial Rings Over Fields

Let F be a field. Let $F[x]$ be the polynomial ring with norm

$$N(p(x)) = \text{degree of } p(x)$$

The ring $F[x]$ is a Euclidean domain. For all $a(x), b(x)$ in $F[x]$ with $b(x) \neq 0$, there exists a unique $q(x), r(x)$ such that

$$a(x) = q(x)b(x) + r(x), \quad N(r(x)) < N(b(x))$$

It follows immediately that $F[x]$ is a PID and UFD.

Gauss's Lemma: Let R be a UFD with field of fractions F . Let $p(x) \in R[x]$. If $p(x)$ is reducible in $F[x]$, then $p(x)$ is reducible in $R[x]$. I.e. if $p(x) = A(x)B(x)$ for $A(x), B(x) \in F[x]$, there exists $r, s \in F$ such that

$$a(x) = rA(x) \in R[x]$$

$$b(x) = sB(x) \in R[x]$$

$$p(x) = a(x)b(x)$$

- This doesn't work for arbitrary integral domains. E.g. $p(x) = x^2 + 1$ is in $\mathbb{Z}[2i] = \{a + 2bi \mid a, b \in \mathbb{Z}\}$. The field of fractions of $\mathbb{Z}[2i]$ is $F = \{a + bi \mid a, b \in \mathbb{Q}\}$. $p(x)$ factors into $(x - i)(x + i)$, thus $p(x)$ is reducible in $F[x]$, but not in $\mathbb{Z}[2i]$.

Theorem: R is a UFD iff. $R[x]$ is a UFD.

8.2 Factoring Polynomials

Theorem: Let $f(x) = a_n x^n + \cdots + a_1 x + a_0$ with $a_i \in \mathbb{Z}$. If $r/s \in \mathbb{Q}$ and r, s are relatively prime, and $p(r/s) = 0$, then $r \mid a_0$ and $s \mid a_n$.

Proof: Substitute.

$$\begin{aligned} a_n (r/s)^n + a_{n-1} (r/s)^{n-1} + \cdots + a_0 &= 0 \\ a_n r^n + a_{n-1} r^{n-1} s + \cdots + a_0 s^n &= 0 \\ a_n r^n &= s(-a_{n-1} r^{n-1} - \cdots - a_0 s^{n-1}) \end{aligned}$$

Thus s divides $a_n r^n$, and since $\gcd(r, s) = 1$, s divides a_n . Similar reasoning shows that r divides a_0 .

- E.g. Showing that $x^3 - 3x - 1$ is irreducible in $\mathbb{Z}[x]$. By Gauss's Lemma, it suffices to show that it is irreducible in $\mathbb{Q}[x]$. By the above theorem, the possible rational roots are ± 1 , neither of which work.
- Thus the polynomial is irreducible.

Sometimes it's helpful to reduce modulo an ideal.

Eisenstein's Criterion: Let P be a prime ideal of integral domain R . Let $f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0$ be a polynomial in $R[x]$. If $a_{n-1}, \dots, a_1, a_0$ are in P but a_0 is not in P^2 , then $f(x)$ is irreducible in $R[x]$.

For $\mathbb{Z}[x]$: Given a prime p , if p divides all a_i in $f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0$ but p^2 doesn't divide a_0 , then $f(x)$ is irreducible in $\mathbb{Z}[x]$ and $\mathbb{Q}[x]$.

- E.g. we know immediately that $x^4 + 10x + 5$ is irreducible in $\mathbb{Z}[x]$ and $\mathbb{Q}[x]$ by testing with $p = 5$.
- Eisenstein's criterion isn't a sufficient condition. E.g. $x^4 + 10x + 25$ works with $p = 5$ but is irreducible.

8.3 Gröbner Bases

A commutative ring R with identity 1 is called **Noetherian** if every ideal of R is finitely generated.

Hilbert's Basis Theorem: If R is Noetherian, so is $R[x]$.

- Since a field F only has finite ideals $\{0\}$ and F , it's Noetherian.
- Thus every ideal in $F[x_1, \dots, x_n]$ is finitely generated as well.

The **Gröbner basis** for ideal I of $F[x_1, \dots, x_n]$ is a finite set of generators $\{g_1, \dots, g_m\}$ whose leading terms generate the ideal of leading terms of I .

$$\begin{aligned} I &= (g_1, \dots, g_m) \\ LT(I) &= (LT(g_1), \dots, LT(g_m)) \end{aligned}$$

- The leading term here is in lexicographic order. E.g. if $x > y$, then $LT(x^3y - xy^2 + 1) = x^3y$.
- Or $LT(y^{67} - x^3y + 2y^4)$ is $-x^3y$.

- The Gröbner bases provide a way to divide polynomials in $F[x_1, \dots, x_n]$ with a **unique remainder**. I.e. the Gröbner bases provide a unique representation for the coset of f in the quotient ring $F[x_1, \dots, x_n]/I$.
- Further, Gröbner bases always exist. **Existence and uniqueness (of remainder)**.

Algorithms exist to find Gröbner bases. Not that big of a topic. Polynomial division is the standard one from middle school, no tricks.